OLIST E-COMMERCE

SQL DATA ANALYSIS PROJECT

A business-focused analysis of orders, customers, products, payments and delivery performance

Project type	SQL-focused portfolio project
Database	MySQL
Domain	E-commerce marketplace
Dataset	Brazilian Olist public dataset
Analysis approach	Data validation, SQL analysis and business interpretation

Prepared as a SQL portfolio project. Findings are based on the recorded analysis results and are subject to the limitations described in this report.

1. Executive Summary

This project analyses historical transactions from the Olist Brazilian e-commerce marketplace dataset using MySQL. It explores order status, recorded item and freight values, product categories, payment methods, customer repeat purchasing, delivery performance and customer reviews.

The analysis found that 96,478 orders had a delivered status in the original status distribution. For delivery-date calculations, eight delivered-status records with invalid zero delivery dates were excluded, leaving 96,470 valid delivered orders; 7,826 of these were late (8.11%). Credit cards were the dominant payment method by record count and recorded payment value. Health & Beauty had the highest item value among the analysed categories. The repeat-customer rate was 3.12% in the available dataset.

The project demonstrates SQL-based data validation, joins, aggregation, conditional calculations, date analysis and business interpretation. Observed associations are not treated as proof of causation.

2. Project Objectives

The objectives were to understand order performance, measure recorded item and freight values, identify leading product categories, examine payment preferences, measure repeat purchasing, investigate late deliveries and explore how delivery outcomes relate to distance, customer state and review scores.

3. Dataset and Database Structure

Table	Purpose
olist_orders_dataset	Order identifiers, status and timestamps
olist_order_items_dataset	Products, sellers, item prices and freight
olist_customers_dataset	Customer identifiers and locations
olist_products_dataset	Product details and category names
olist_order_payments_dataset	Payment methods, values and instalments
olist_order_reviews_dataset	Review scores and customer comments
olist_sellers_dataset	Seller identifiers and locations
olist_geolocation_dataset	Coordinates associated with ZIP prefixes
product_category_name_translation	Portuguese-to-English category mapping

Key relationships used in analysis: Orders to Customers via customer_id; Order Items to Orders via order_id; Order Items to Products via product_id; Order Items to Sellers via seller_id; Payments and Reviews to Orders via order_id; Products to category translation via the category name. These are analytical join relationships; this report does not claim that formal database foreign-key constraints were created.

4. Data Validation and Quality Checks

The imported tables were checked for row counts, identifier uniqueness, missing values, unmatched records and invalid date values. The Orders table was initially imported twice; the duplicate import was corrected by reloading it once. The final Orders count was 99,441 rows and 99,441 distinct order IDs.

Table	Rows
Orders	99,441
Order Items	112,650
Customers	99,441
Products	32,951
Payments	103,886
Reviews	99,223
Sellers	3,095
Geolocation	1,000,163
Category Translation	71

Additional validation findings

• Orders, Customers, Products and Sellers had row counts matching their distinct identifier counts.
• Order Items, Payments and Reviews have different row grains: one order can have multiple items, payment records or reviews. These tables must be aggregated carefully to avoid inflated counts and values.
• Required order-item fields checked during validation had no missing values, and unmatched order-item-to-order and order-to-customer checks returned zero.
• One delivered order had no corresponding payment record; it was retained as an exception rather than assigned an assumed explanation.
• Eight delivered-status records contained zero delivery dates and were excluded from valid delivery-date calculations.
• 610 products had blank category names; these appeared in 1,603 order-item records and represented R$179,619 in item value.
• Seven seller ZIP prefixes and 157 distinct customer ZIP prefixes did not match the geolocation ZIP-prefix table.

Data-quality handling was analytical: exceptions were identified and documented rather than silently deleted or given invented values.

5. SQL Analysis and Results

5.1 Order status

The Orders table contained 99,441 records. Delivered status accounted for 96,478 orders (97.02%); canceled status accounted for 625 orders (0.63%). The remaining records were in shipped, unavailable, invoiced, processing, created or approved statuses.

5.2 Recorded item and freight values

Metric	Result
Total item value	R$13,596,989.00
Total freight value	R$2,251,909.54

Combined item and freight value	R$15,848,898.54
Freight share of combined value	14.21%

These are recorded item and freight values, not profit. Freight collected from customers is not necessarily equal to the marketplace's shipping expense.

5.3 Product categories

Health & Beauty had the highest recorded item value at R$1,258,951, followed by Watches & Gifts (R$1,205,213), Bed, Bath & Table (R$1,037,729), Sports & Leisure (R$988,433) and Computers & Accessories (R$912,344). Blank categories represented R$179,619 in item value.

5.4 Payment methods

Payment method	Records	Recorded value
Credit card	76,795	R$12,542,084.19
Boleto	19,784	R$2,869,361.27
Voucher	5,775	R$379,436.87
Debit card	1,529	R$217,989.79
Not defined	3	R$0.00

Credit cards represented approximately 73.92% of payment records and around 77.5% of recorded payment value. Payment counts refer to payment records, not unique orders.

5.5 Customer repeat purchasing

Metric	Result
Unique customers	96,096
Repeat customers	2,997
One-time customers	93,099
Repeat-customer rate	3.12%

The repeat-customer rate describes the historical dataset period and should not be treated as a current retention metric without a more recent observation period.

5.6 Delivery performance

Metric	Result
Valid delivered orders	96,470
Late orders	7,826
On-time or early orders	88,644
Late-delivery rate	8.11%
On-time or early rate	91.89%
Average purchase-to-delivery time	12.54 days
Average processing time	2.78 days
Average transit time	9.31 days

The average processing time is measured from approval to carrier handover; transit time is measured from carrier handover to customer delivery. The average purchase-to-delivery time uses purchase timestamp to customer delivery. These averages use valid available timestamps and are not necessarily based on identical subsets.

5.7 Delivery distance

ZIP-prefix average coordinates were used to estimate seller-to-customer straight-line distance with the Haversine formula. The analysis included 97,325 seller-order records with both seller and customer coordinates.

Estimated distance	Records	Late rate
Under 100 km	18,102	6.32%
100–499 km	37,559	7.07%
500–999 km	26,134	8.31%
1,000–1,999 km	9,873	10.68%
2,000 km and above	5,657	13.61%

Late-delivery rates increased across the longer distance groups. This is an association, not proof that distance alone causes delay. ZIP-prefix average coordinates are approximations and do not represent actual road or courier routes. Seller-order records are not the same as unique orders because one order may involve multiple sellers.

5.8 State-wise delivery performance

Among states with at least 100 valid delivered orders, AL had a late rate of 23.93% (95 of 397 orders), while SP had 2,387 late orders among 40,494 valid delivered orders (5.89%). A high late-order count can reflect high order volume; percentages and sample sizes must be considered together.

5.9 Review scores and delivery performance

On-time or early orders had 88,660 review records with an average score of 4.29. Late orders had 7,700 review records with an average score of 2.57. One-star review shares were 6.60% for on-time or early orders and 46.16% for late orders. These results show an association, not proof of causation. Review-record counts differ from delivered-order counts because orders do not all have exactly one review.

6. Overall Business Insights

1. Most orders in the dataset had delivered status.
2. Recorded item value was approximately R$13.60 million, with R$2.25 million in freight value.
3. Health & Beauty led the analysed product categories by item value.
4. Credit cards dominated payment records and recorded payment value.
5. Repeat customers represented 3.12% of unique customers in the dataset.
6. The late-delivery rate among valid delivered orders was 8.11%.
7. Longer estimated distance groups had higher late-delivery rates.
8. Late deliveries were associated with lower customer review scores.

7. Business Recommendations

• Investigate customer retention and repeat-purchase opportunities.
• Review delivery performance by state, seller and distance group, accounting for order volumes.
• Examine long-distance logistics and whether delivery estimates reflect operational realities.
• Review negative customer feedback associated with late deliveries.
• Improve product-category completeness to support more reliable category reporting.
• Monitor payment experience around the payment methods most frequently used.

These are proposed areas for further investigation. The analysis does not establish that any single intervention will cause an improvement.

8. Limitations

• The dataset is historical and does not represent live marketplace performance.
• One delivered order had no matching payment record; no explanation was assumed.
• Eight delivered-status records had invalid zero delivery dates and were excluded from valid delivery-date analysis.
• Some customer and seller ZIP prefixes did not match the geolocation table.
• ZIP-prefix average coordinates approximate location; Haversine distance is straight-line distance, not road distance.
• The Orders table does not contain a carrier identifier, so carrier-specific performance could not be measured.
• Relationships between distance, delivery and reviews are observational and do not establish causation.

9. Conclusion

This project demonstrates how SQL can be used to explore a multi-table e-commerce database, validate data, join related tables, calculate business metrics and communicate findings. The work highlights the importance of understanding table grain, avoiding duplicate counting, documenting data-quality exceptions and interpreting analytical results carefully.

Portfolio skills demonstrated: MySQL, SQL querying, joins, aggregation, subqueries, CASE expressions, data validation, date calculations, customer analysis, business interpretation and analytical communication.